SMARTX: Porewater Ammonium

Septmeber 2025 Samples

2026-02-23

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##Run Information

```r
cat("Run Information:  ") #lets you know what section you're in
```

```
## Run Information:
```

```r
#set the run date & user name
  sample_year <- "2025"
  sample_month <- "September"
  user <- "Brenden Blakley"

#identify the files you want to read in
  #read in as a list to accommodate multiple runs in a month
  NH3_files <- c("Raw Data/20260213_SMARTX_Sept2025_Run1.csv",
                 "Raw Data/20260217_SMARTX_Sept2025_Run2.csv",
                 "Raw Data/20260218_SMARTX_Sept_2025_Run3.csv")

# Define the file path for QAQC log file - NO Need to change just check year
  file_path <- "Raw Data/SEAL_SMARTX_QAQC_Log_2025.csv"
  final_path <- "Processed Data/GCReW_SMARTX_Porewater_NH4_202509.csv"

#record any notes about the run or anything other info here:
  run_notes <- ""

#Add SMARTX Prefix if necessary to samples
  Add_Prefix <- "YES"

  #Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

  cat(run_notes)
```

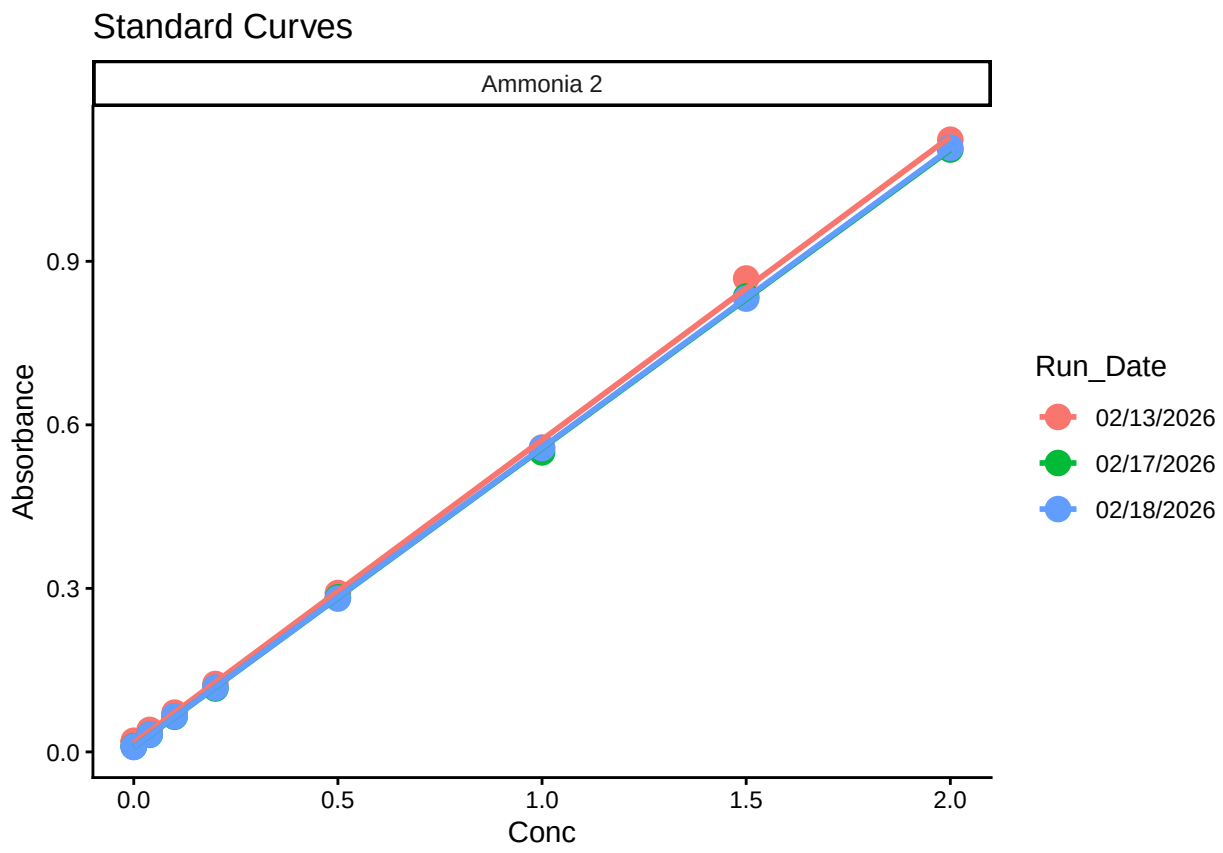
##Setup

##Read in metadata and create similar sample IDs for matching to samples

0.1 Import Data & Clean

0.2 Assessing standard Curves

1 Assess Standard Curves

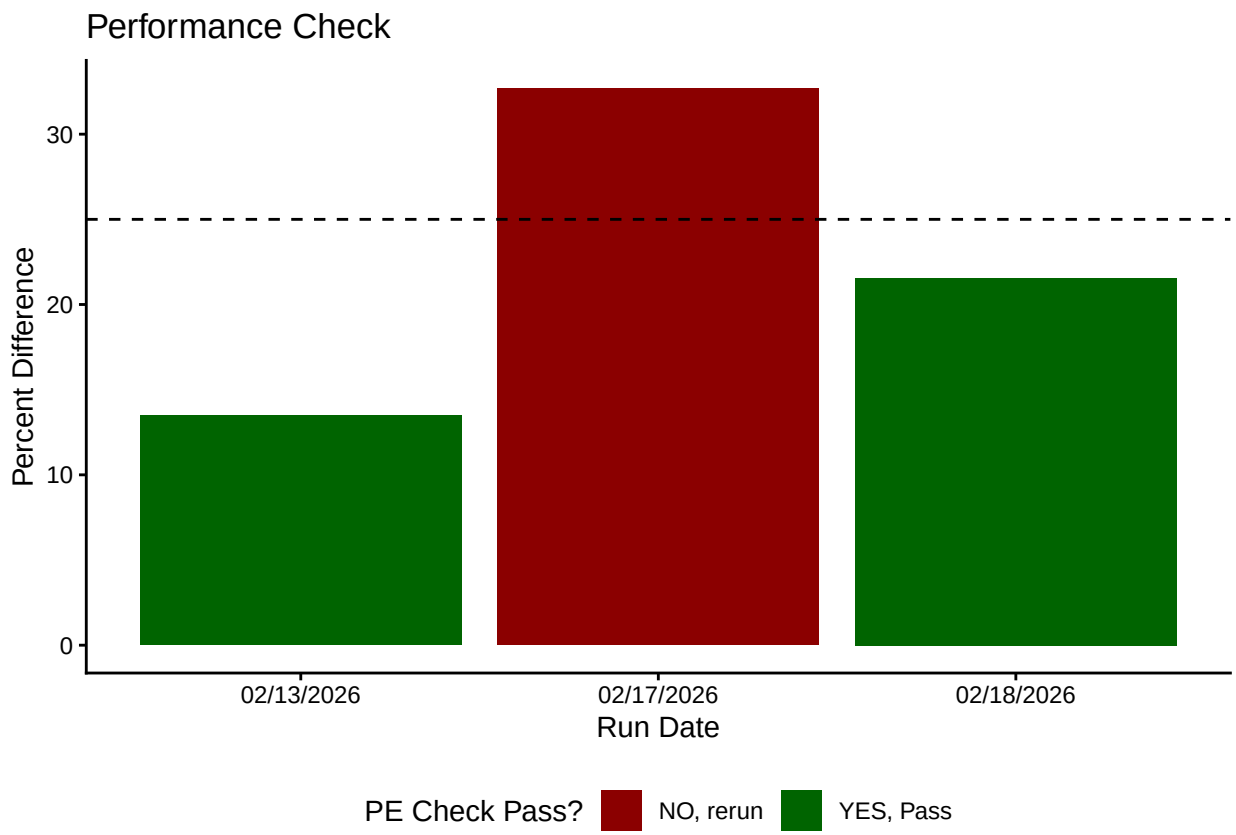


1.1 Dilution Corrections - ensure the latest dilution is kept

#Dilutions

1.2 Performance Check

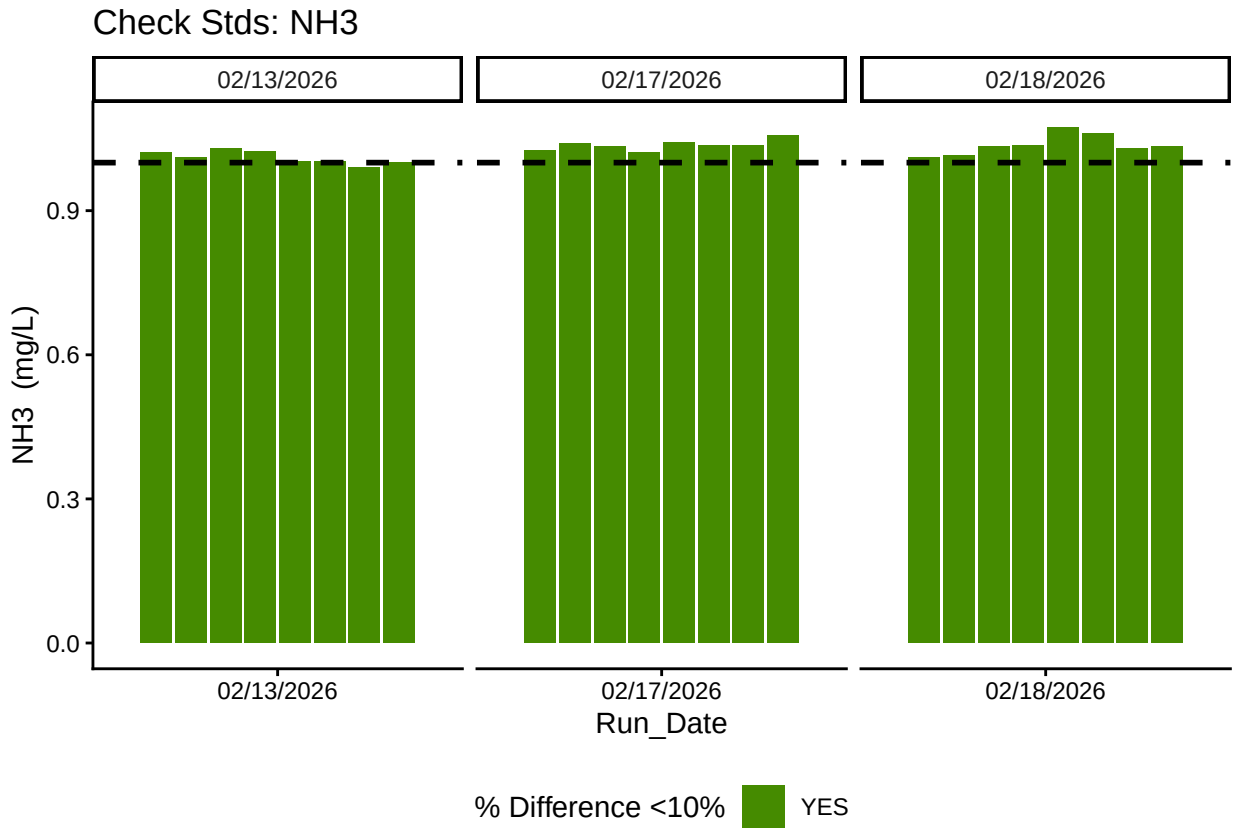
```
## [1] "02/13/2026--NH3 pe Check has a % Difference <25% - PROCEED"
## [2] "02/17/2026--NH3 pe Check has a % Difference >25% - REASSESS"
## [3] "02/18/2026--NH3 pe Check has a % Difference <25% - PROCEED"
```



1.3 Analyze the Check Standards

```
## Analyze Check Standards
```

```
## # A tibble: 3 x 1
##   RSD_flag
##   <chr>
## 1 02/13/2026--NH3 Check Standard RSD within Range - PROCEED
## 2 02/17/2026--NH3 Check Standard RSD within Range - PROCEED
## 3 02/18/2026--NH3 Check Standard RSD within Range - PROCEED
```




```
## # A tibble: 3 x 1
##   CHK_60_Flag
##   <chr>
## 1 02/13/2026-- >60% of NH3 Check Standards are within range of expected concent~
## 2 02/17/2026-- >60% of NH3 Check Standards are within range of expected concent~
## 3 02/18/2026-- >60% of NH3 Check Standards are within range of expected concent~
```

1.4 Analyze Blanks

```
## Assess Blanks
```

```
## # A tibble: 3 x 1
##   BLK_60_Flag
##   <chr>
## 1 02/13/2026-- >60% of NH3 Blanks are below the lower 25% quartile of samples -~
## 2 02/17/2026-- >60% of NH3 Blanks are below the lower 25% quartile of samples -~
## 3 02/18/2026-- >60% of NH3 Blanks are below the lower 25% quartile of samples -~
```

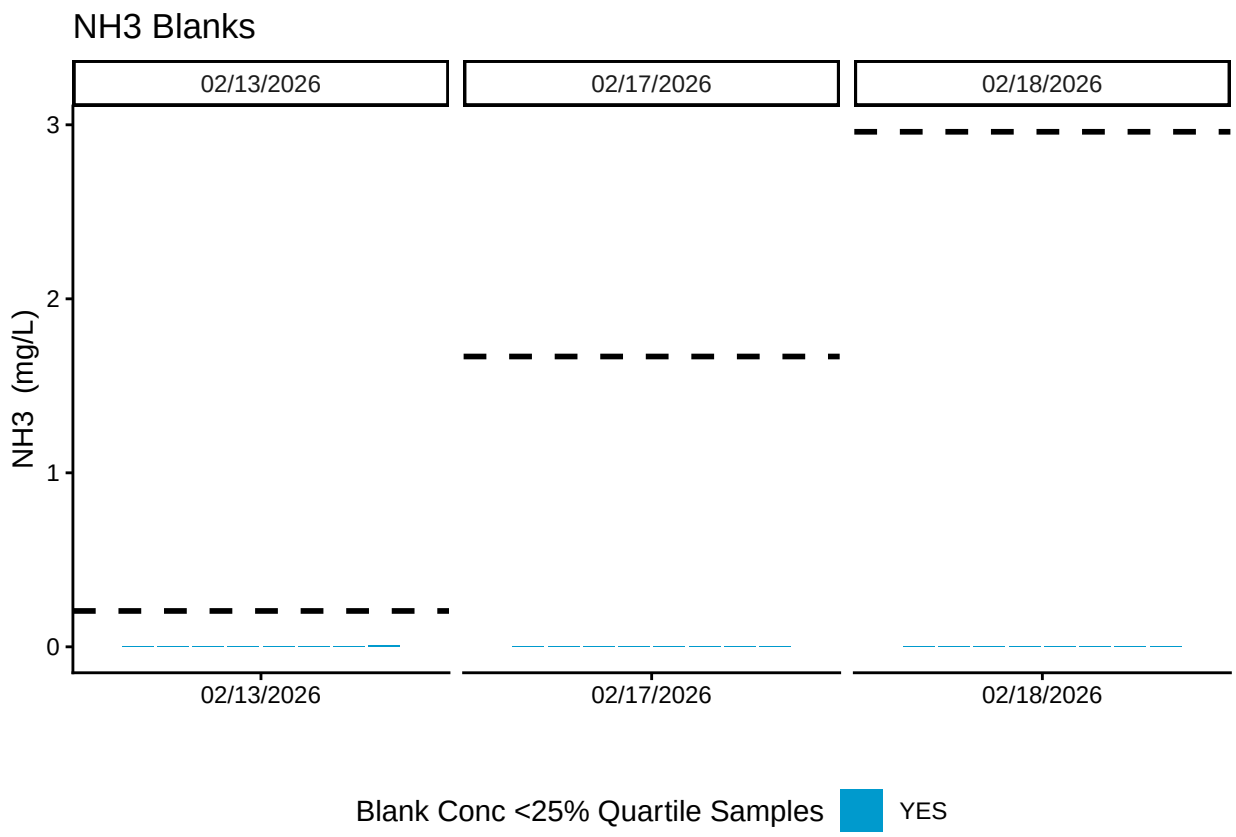
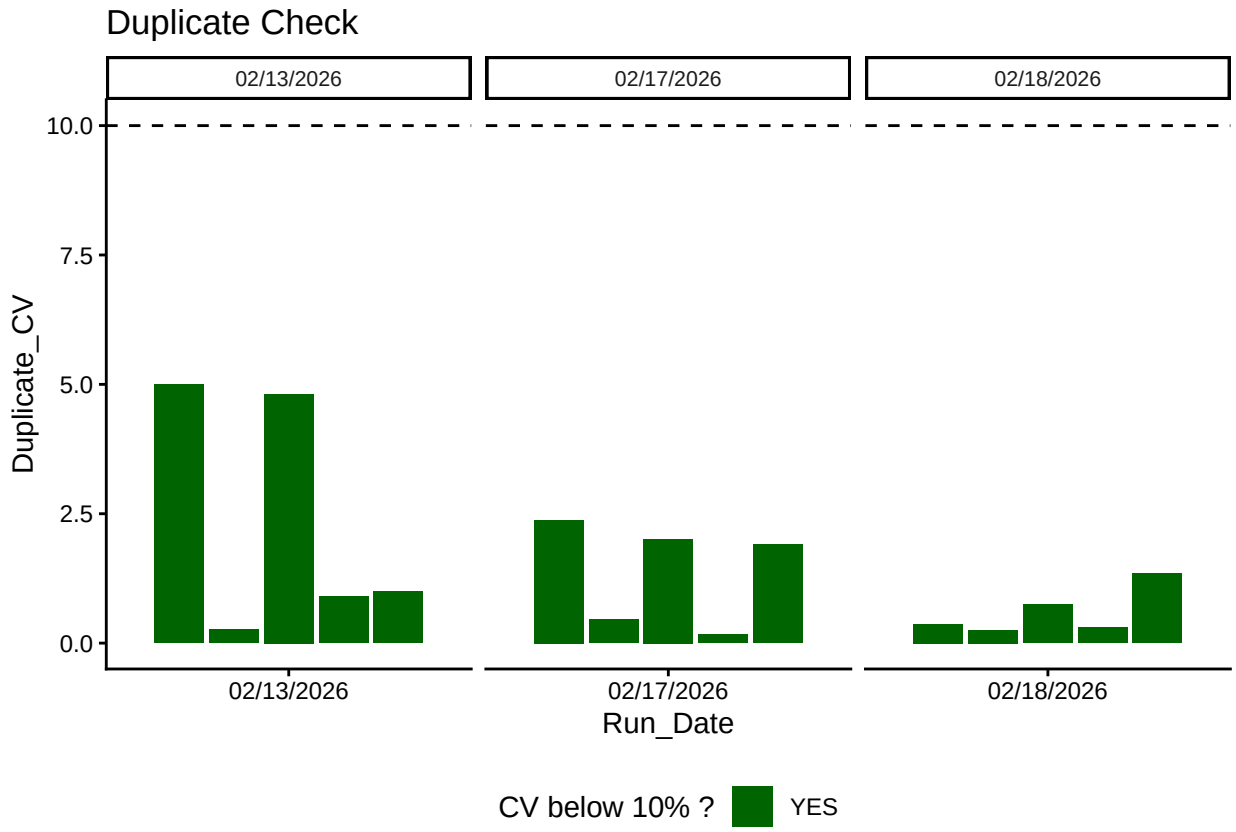


Table 1: Mean Concentration of Blanks

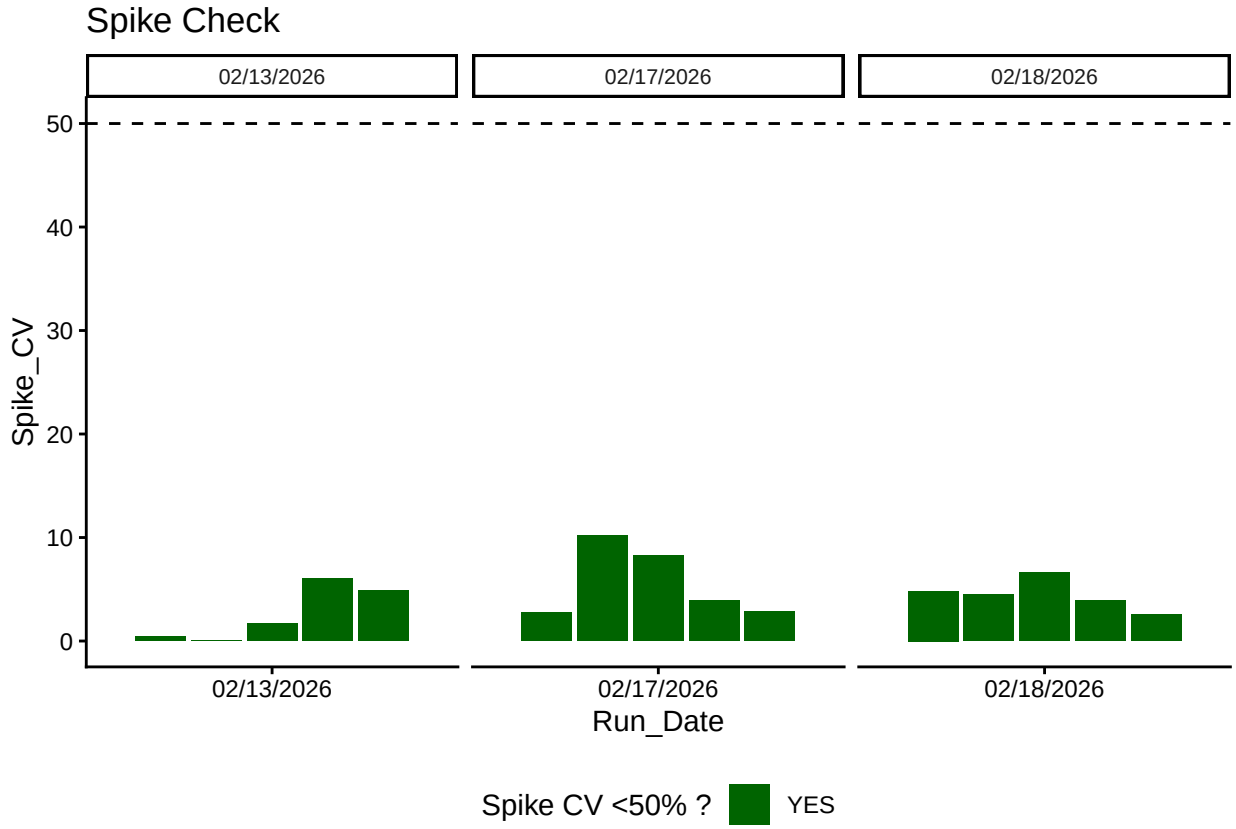
Test	Blank_Mean_Conc
NH3	0.0027

1.5 Analyze Duplicates

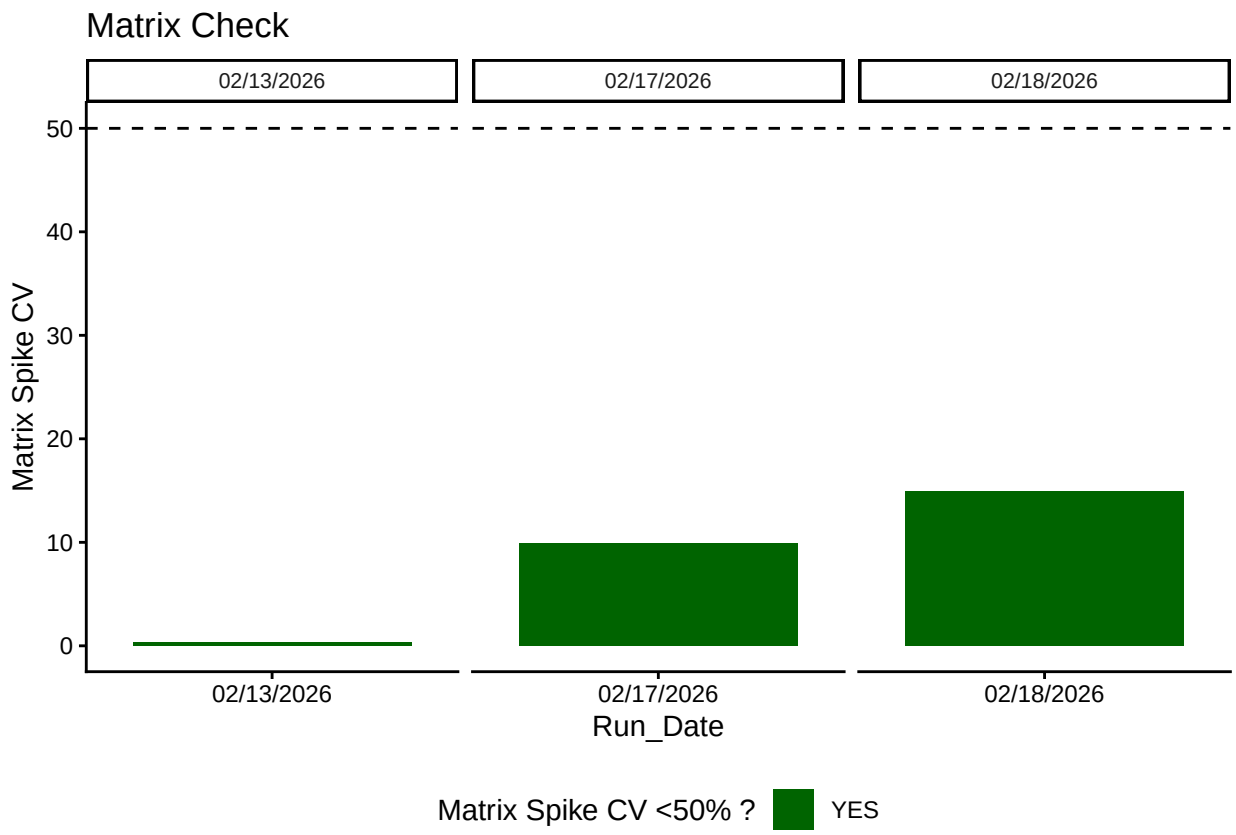
```
## Analyze Duplicates
```



1.6 Spikes


Spike Check
02/13/2026
02/17/2026
02/18/2026
50
40
30
20
10
0
Spike_CV
02/13/2026
02/17/2026
02/18/2026
Run_Date
Spike CV <50% ?
YES

1.7 Matrix Effects



1.8 Unit Converted Data Column Added (mg/L to uM)

1.9 Sample Flagging - Within range of standard curve

```
## Sample Flagging
```

1.10 Add QAQC Flags

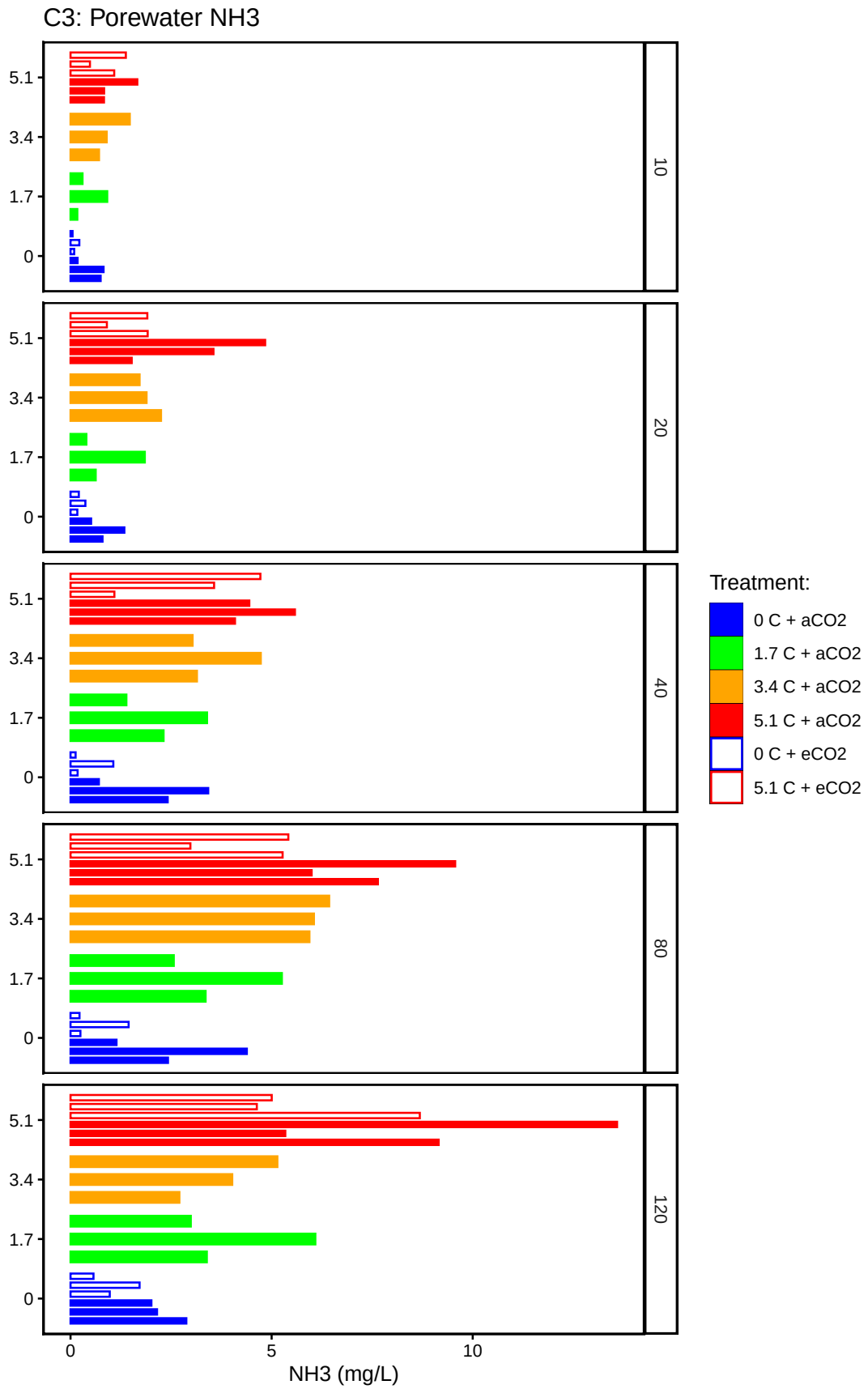
1.11 Merge samples with Metadata & check all samples are present

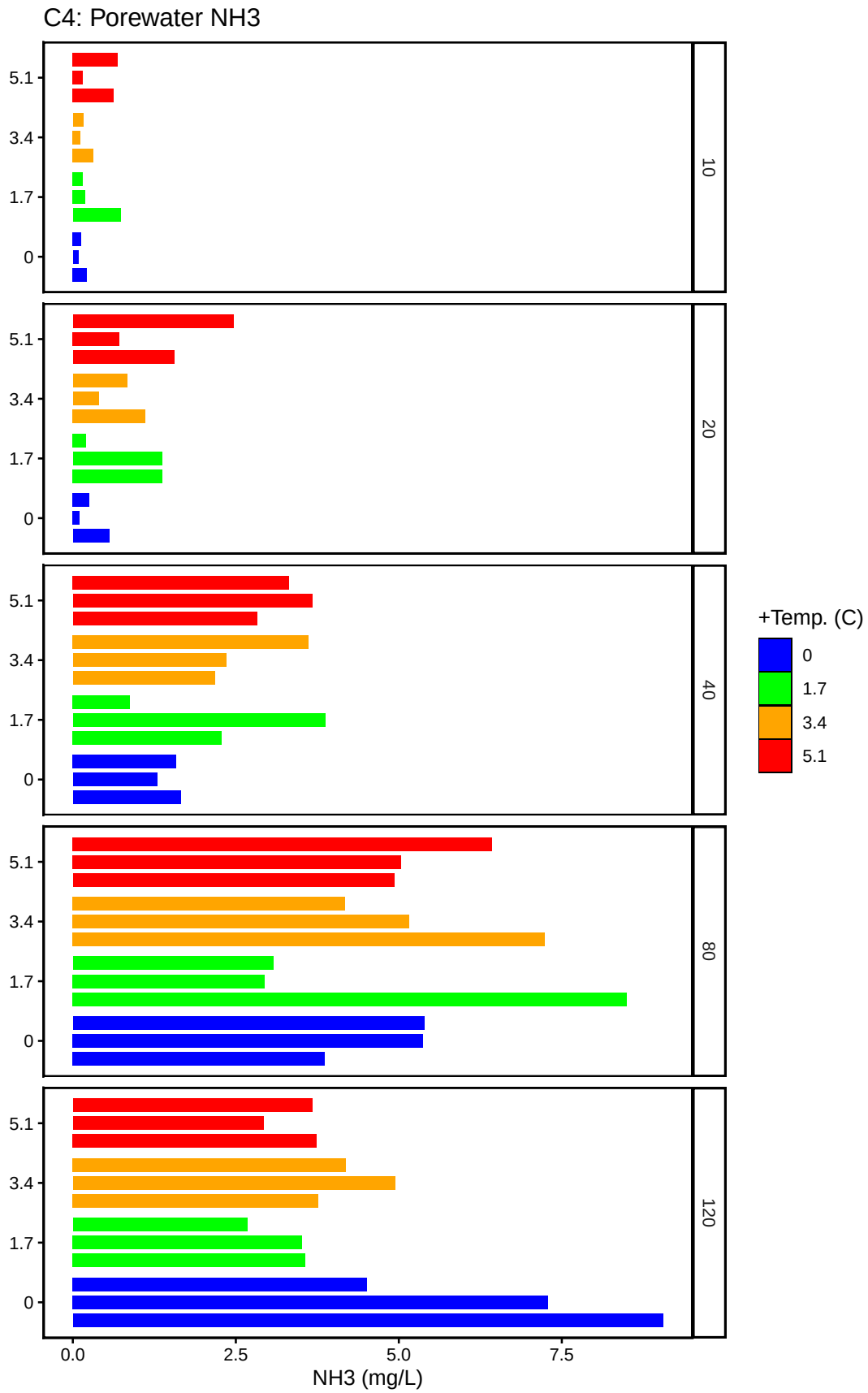
```
## Merge Samples with Metadata
```

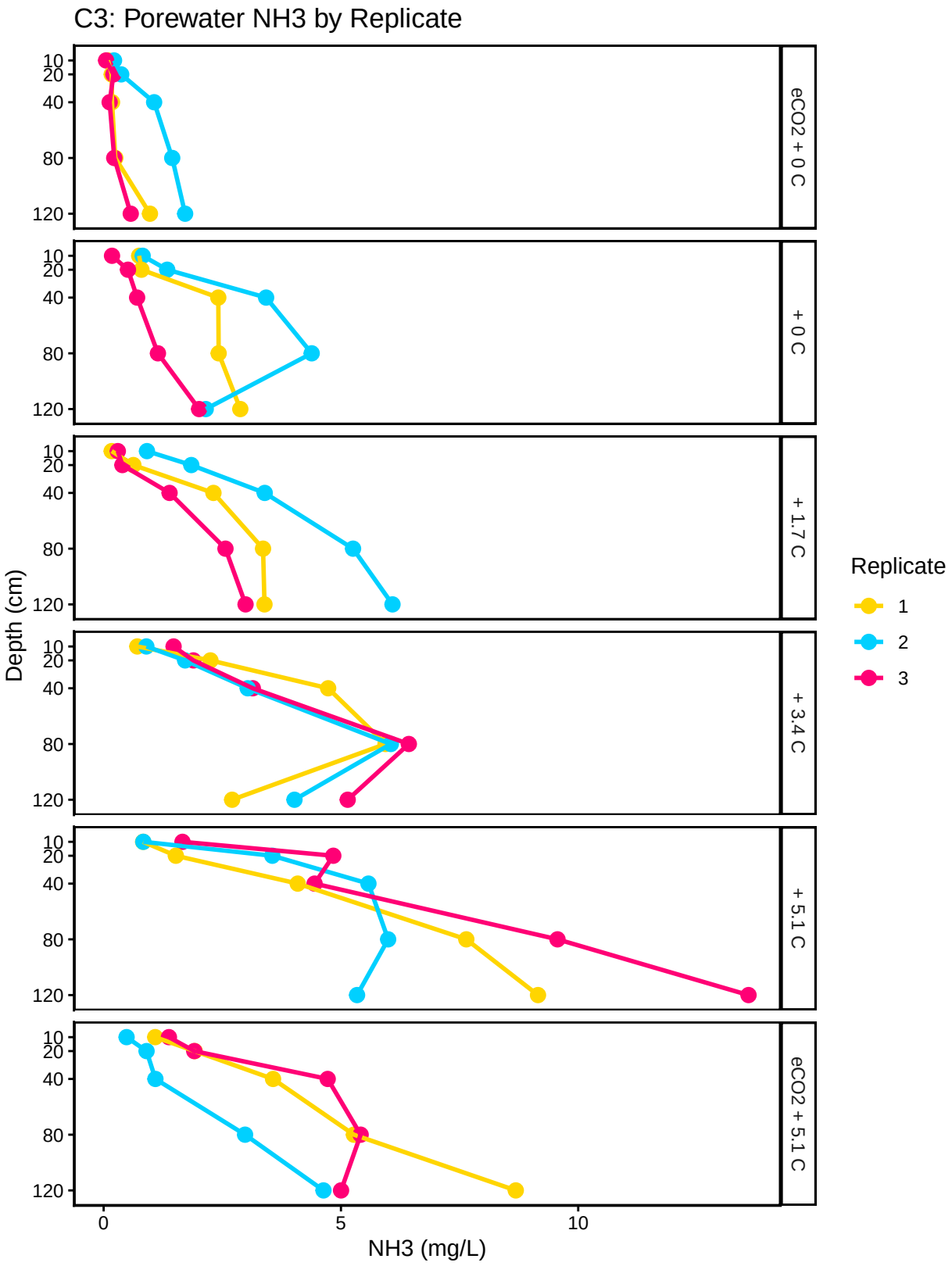
```
## All sample IDs are present in data
```

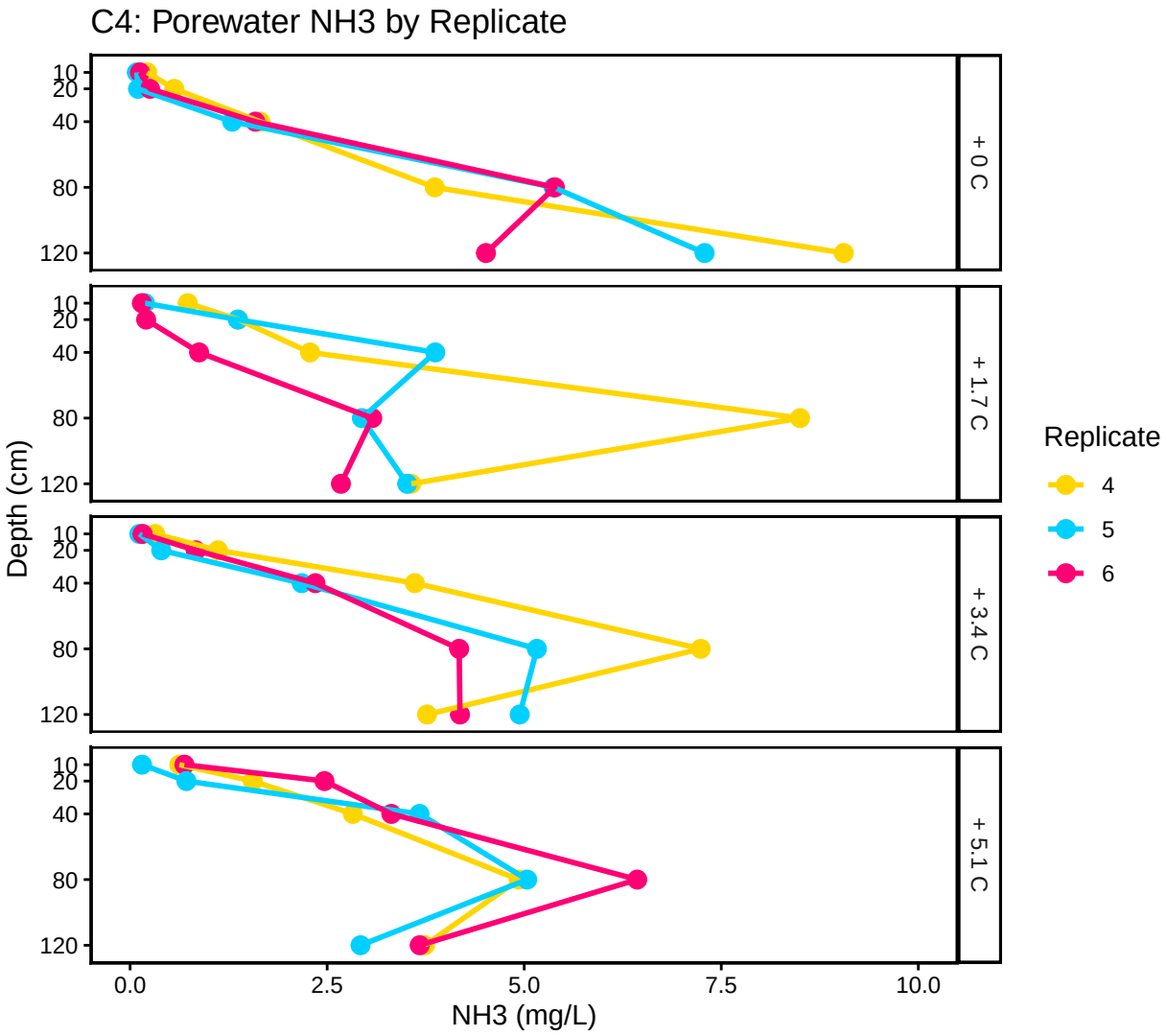
1.12 Visualize Data

```
## Visualize Data
```


C3: Porewater NH3
5.1
3.4
1.7
0
10
20
40
80
120
0
5
10
NH3 (mg/L)
Treatment:
0 C + aCO2
1.7 C + aCO2
3.4 C + aCO2
5.1 C + aCO2
0 C + eCO2
5.1 C + eCO2


C4: Porewater NH3
5.1
3.4
1.7
0
10
20
40
80
120
0.0
2.5
5.0
7.5
NH3 (mg/L)
+Temp. (C)
0
1.7
3.4
5.1


C3: Porewater NH3 by Replicate
Depth (cm)
NH3 (mg/L)
eCO2 + 0 C
+ 0 C
+ 1.7 C
+ 3.4 C
+ 5.1 C
eCO2 + 5.1 C
Replicate
1
2
3



1.13 Organize

#Export Data

#end